

Project Design Phase Solution Architecture

Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

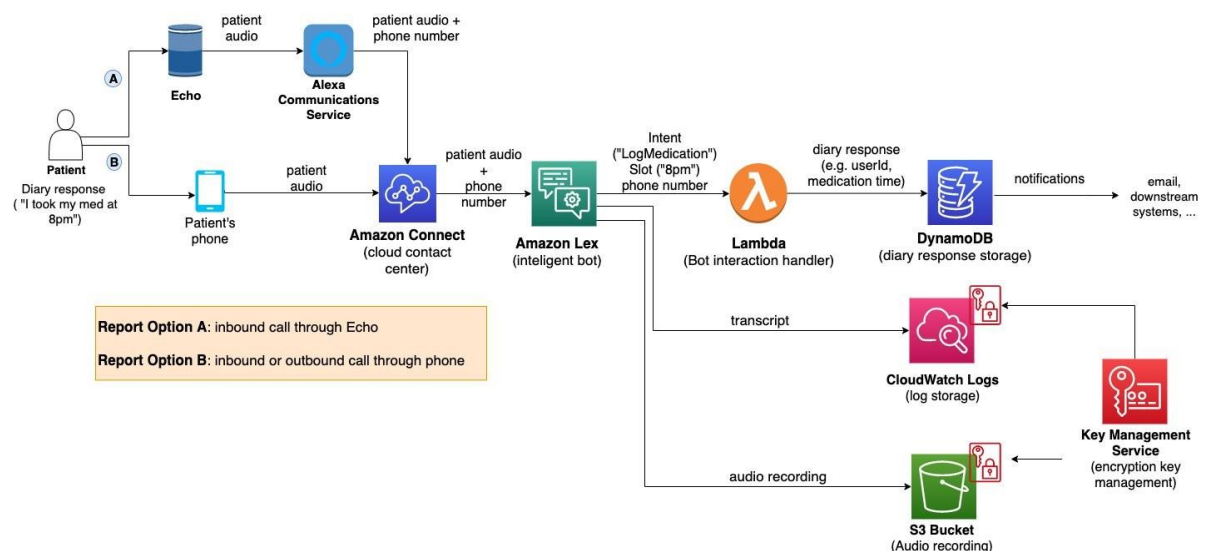
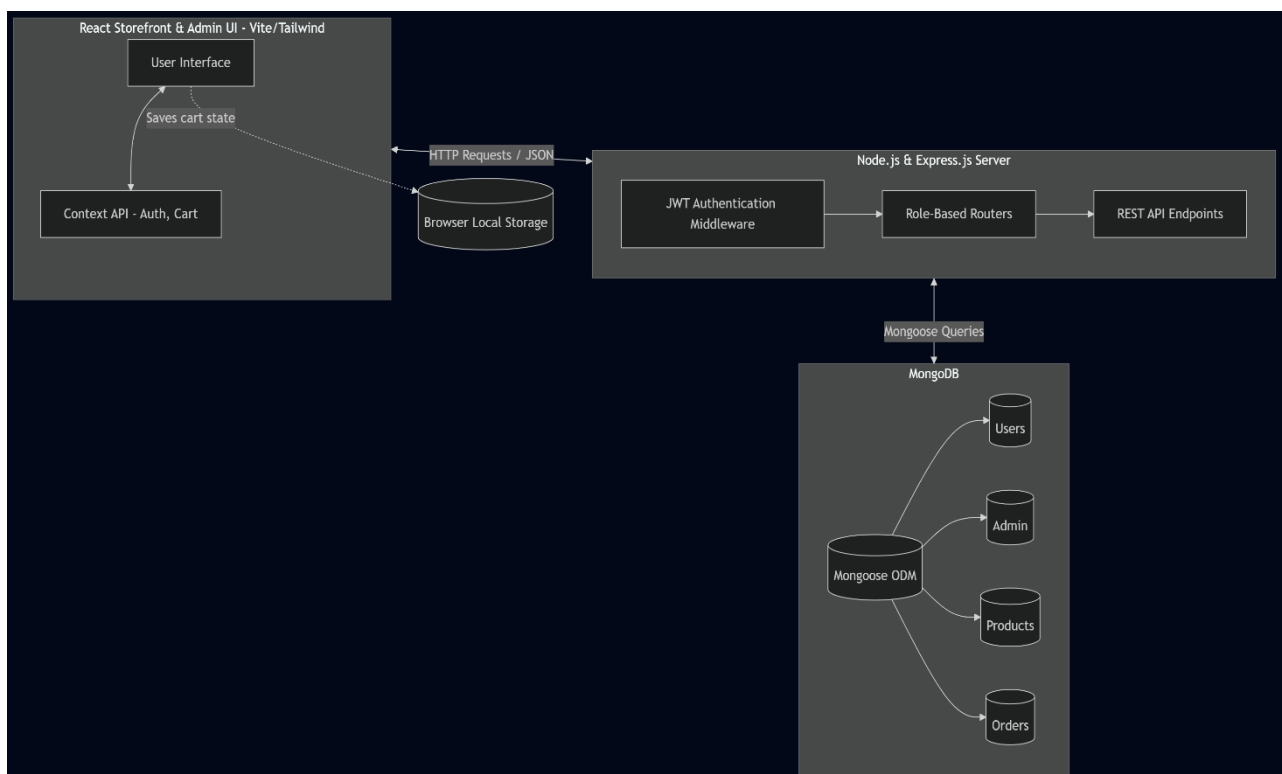


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

Solution Architecture Description: The application implements a decoupled MERN stack architecture. The frontend uses React and Tailwind CSS, utilizing the Context API for global states like authentication, carts, and wishlists. The Node/Express backend serves as a stateless RESTful API, secured by JWT validation middleware, to handle logic for products, users, and orders. The database layer utilizes MongoDB and Mongoose ODM for data storage, relationship modeling, and validation.



Entity-Relationship Diagram (Database Schema):

